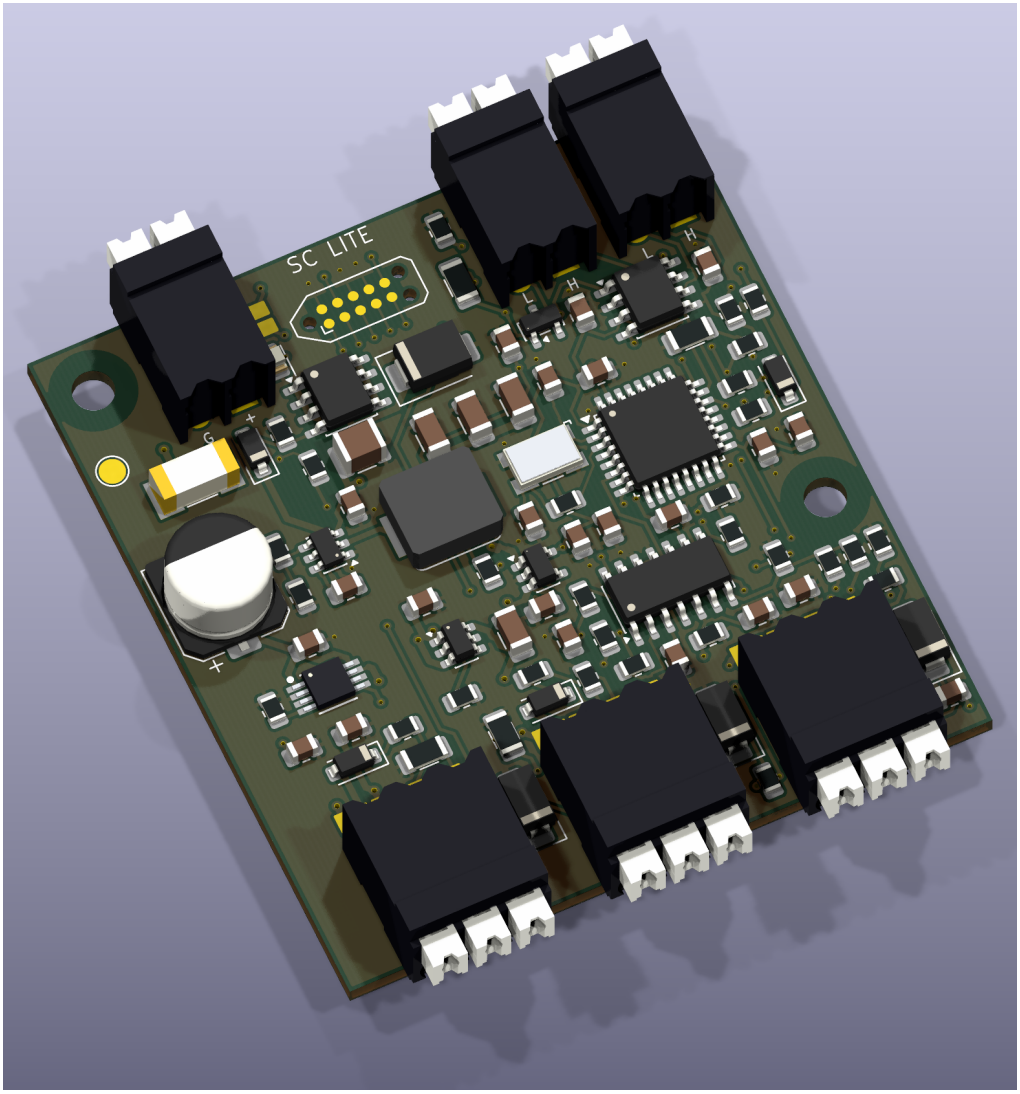
FRC
Sensor CAN

Revision C Lite · Prototype engineering and assembly guide

Three industrial / logic sensor inputs
Protected battery and regulated 5 V sensor power



2026-09-08

Editable KiCad files, DigiKey BOM, firmware, robot examples and verification records accompany this guide. No physical prototype has been tested. Electrical ratings are design targets pending the included qualification plan.

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1 Product overview

Source: README.md

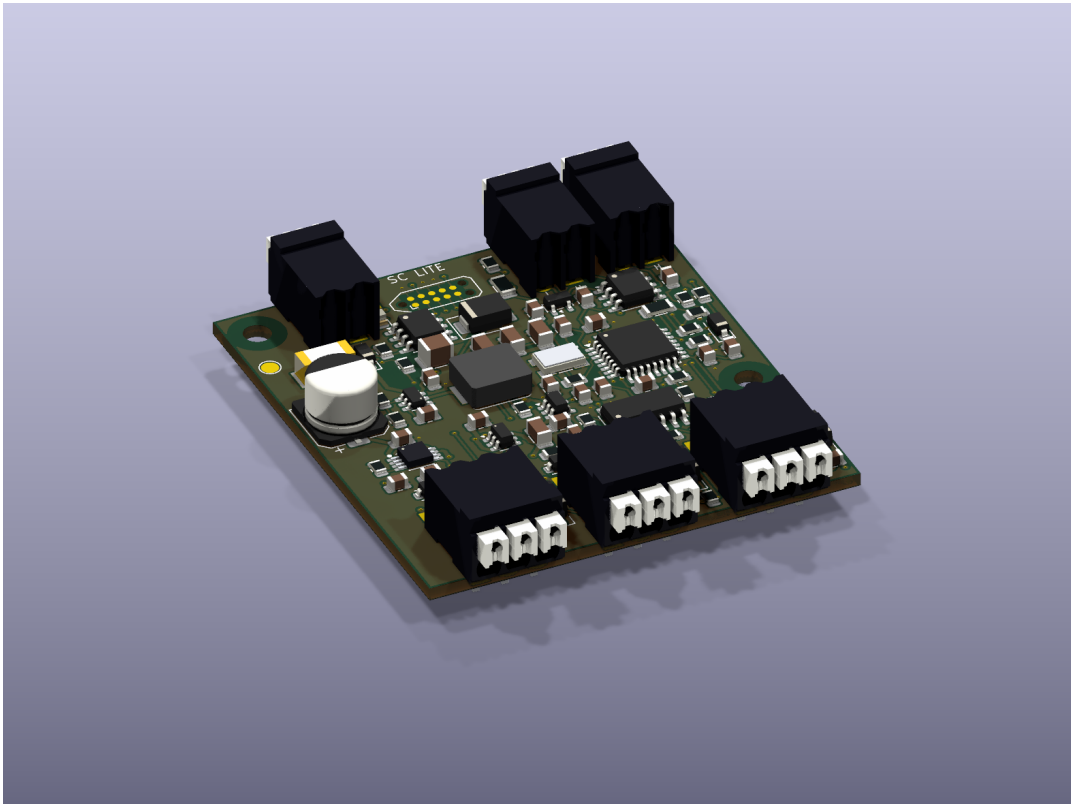
Three digital sensor inputs over CAN, with protected battery and regulated 5 V sensor power. **52 × 56 mm, 89 fitted parts, no plug-in jumpers.** This is the simplified successor to B/B-2; all are unbuilt prototypes.

	Rev B	Rev C Lite
PCB	60 × 68 mm	**52 × 56 mm; 28.6% less area**
Fitted parts	144	**89; 38.2% fewer**
Resistors / capacitors	63 / 34	**33 / 29**
Sensor inputs	3	3
Supply target	5-22 V	**6-22 V**
Sensor power	Software control and monitoring	Hardware protected, always enabled
Configuration	10 headers and loose shunts	7 solder selectors, no purchased jumpers
Input threshold	Logic / industrial selectable	Fixed, 3.3/5 V and industrial NPN/PNP
Dated quantity-one parts estimate	$59.30	**$43.90**

Both designs use horizontal Weidmüller **1824420000** and **1824430000** push connectors. Rev C retains 0805 passives, a leaded LQFP MCU, SOIC comparator/transceiver and compact TC2050 programming pads. The battery current limiter still needs paste/heatbed for its exposed pad. Connector overhang gives a conservative **52 × 64.1 mm assembly envelope**, before wire/button clearance.

Build and review

- Complete prototype package, handbook.
- Editable KiCad project, schematic PDF, interactive assembly BOM.
- DigiKey BOM, DigiKey import, factory instructions.
- Wiring, solder settings and pogo map, assembly, firmware, protocol, robot examples.
- What was removed and why, calculations and qualification, connector evidence, tools.



Operating targets

Common-ground positive 0-30 V signals; LOW ≤1.2 V, HIGH ≥2.4 V at the input, under the documented component/leakage/temperature budgets. 1.8 V recognition is not promised. Each sensor chooses 5 V or protected input voltage with solder pads; **300 mA total per rail across all three ports**. Battery output follows the supply and cannot keep a 12 V sensor running through deep battery sag.

Removing the second converter means regulated 5 V requires at least 6 V input. Removing rail monitoring means CAN cannot distinguish a valid inactive sensor from a sensor that lost power. Sensor outputs are power supplies for sensors only, with no actuator control. Correct a short and cycle input power to reset a latched rail.

The standalone hardware checks, native ERC/DRC, independent exported-copper checks and firmware/driver tests are provided. **No physical board has been assembled or tested.** Startup, sensor loads, transients, EMC, thermal performance and FRC use require the prototype qualification. This original Team Use protocol is not compatible with CTRE Phoenix/Tuner. Prices are dated observations, excluding PCB, assembly, attrition, freight and taxes.

All revisions · Sources and notices

2 Simplification

Source: docs/SIMPLIFICATION.md

The board is now a three-channel digital input interface. The reduction comes from deleting functions, not removing arbitrary bypass components.

Change from B/B-2	Result / tradeoff
Delete the 4.2 V buck + boost arrangement; use one 5.067 V nominal buck	Removes a converter, inductor, diode and associated capacitors; supply minimum becomes 6 V
Delete rail ADC dividers, software enables and fault-report networks	Power is standalone; CAN cannot report sensor supply failure or cycle a sensor
Replace plug-in selectors with solder pads	No ten purchased headers or loose shunts; settings change with an iron
Delete selectable industrial threshold	One reference supports 3.3/5 V and typical NPN/PNP sensors; less noise margin than B's industrial mode
Delete onboard 3.3 V input bias choice	External 3.3 V pullup needed for non-5-V-tolerant open-drain devices
Use one 220 kΩ input series resistor instead of two 110 kΩ	Three fewer resistors; loses the previous single-series-resistor-short fault tolerance
Keep one status LED, remove the other indicator networks	Fewer parts; inspect input state over CAN
Two mounting holes, smaller board and fewer test pads	52 × 56 mm; use the documented mounting and service contacts

Totals: **144 →89 fitted components**, **63 →33 resistors**, **34 →29 capacitors**, **60 ×68 →52 ×56 mm**. All 89 are placed from the top; 83 are SMT and six are through-hole/THR terminals. Solder selectors, pogo contacts, one test point and mounting holes add PCB features, not purchased components.

Why 29 capacitors remain

Capacitors at different locations perform different jobs. Sharing a distant capacitor does not provide the same local current path. MCU supply pins, the comparator and both CAN supply pins retain local bypass. The clock needs its load capacitors, each input needs its own filter, the buck needs its bootstrap capacitor and input/output reservoirs, and the protected rails need local stability/transient capacitance. The battery limiter's 33 nF capacitor sets its latch delay.

Three 22 µF capacitors remain at the 5 V buck output. Their nominal 66 µF becomes approximately **25.9 µF** under the recorded Samsung typical DC-bias curve and additional design budgets; two would leave about **17.3 µF**, below the 25 µF design target. They are not interchangeable with one arbitrary capacitor labelled 47 or 68 µF. A future single-capacitor substitution needs a specific part's bias curve, loop/transient review and physical tests.

The 5 V feedback divider uses 0.1%, 25 ppm/°C resistors to preserve voltage margin. The PNP pulldowns stay 1206 / 0.25 W because a 30 V input dissipates about 90 mW continuously. Other signal resistors remain hand-friendly 0805. The NPN pullup diodes prevent an externally driven signal from feeding its selected sensor rail through the pullup resistor.

Both current-limit ICs, reverse-polarity MOSFET, TVS parts and CAN default-state resistors remain. Removing those would save little area compared with the terminals while weakening behavior under miswiring, faults or reset. The MCU includes a CAN controller, but still needs the physical CAN transceiver. An ESP32-C3 would not eliminate that transceiver and would add module/firmware considerations without a clear assembly advantage here.

3 Wiring and programming contacts

Source: docs/WIRING.md

Use Rev C hardware, firmware and host tools together. Board coordinates below are viewed from the component side, origin at the top-left PCB corner. Bottom silkscreen is read after turning the board over. Do not infer pin numbering from left/right after flipping.

Connector	Pin 1	Pin 2	Pin 3
J1 power	Positive input, 6-22 V	GND	-
J2, J3 CAN	CAN H	CAN L	-
J4 input 1	Selected sensor power	GND	Signal
J5 input 2	Selected sensor power	GND	Signal
J6 input 3	Selected sensor power	GND	Signal

Each terminal pole has two solder legs; they are one electrical contact. J2/J3 are connected in parallel for bus continuation, with no onboard CAN ground terminal. Common ground comes through robot power. Use twisted H/L wiring; keep branches short. Strip 8 mm using the conductor range specified by Weidmüller for the exact terminal. Press the white release button for removal and for conductor types requiring it. Provide cable strain relief independently of the terminal.

Feed J1 from appropriately fused robot distribution. F1 is secondary board protection. Do not backfeed J4-J6 power pins, parallel the two sensor rails, or apply external power to the programming rail. The board is not galvanically isolated.

Solder settings

All selectors ship **open**, with no stencil paste on their pads. Configure with power disconnected, then inspect and check continuity. A small iron bridge joins adjacent pads. Remove solder before changing settings.

Channel	Power selector	Bias selector	Center X, mm
1 / J4	JP101	JP103	Power 7; bias 13
2 / J5	JP201	JP203	Power 23; bias 29
3 / J6	JP301	JP303	Power 39; bias 45

These six selectors are at Y=43.5 mm, just above the sensor connectors. **Pads 1, 2, 3 run left to right when viewed from the component side.** The square pad identifies pin 1. Use the detailed assembly drawing/interactive BOM to identify a selector before soldering.

Selector	Bridge 1-2	Bridge 2-3	Open
Power	Regulated 5 V	Protected battery/input voltage	Port power disconnected
Bias	10 kΩ pullup to selected port supply, through diode: NPN/open collector	10 kΩ pulldown: PNP	Push-pull or external pullup

One bridge per selector. Bridging both sides of a power selector shorts battery to 5 V. Bridge JP2 only at a CAN bus end that needs a 120 Ω termination. Leave it open at intermediate nodes; normally there are two terminations on the entire bus.

Examples:

• Industrial NPN: select the sensor's required supply; bias 1-2. Output current from the pullup is below about 2.2 mA at 22 V. Select inversion in firmware if asserted means pulled low.

• Industrial PNP: select supply; bias 2-3. Verify the sensor's OFF leakage cannot raise the input above the LOW target through 10 kΩ (1.2 V / 10 kΩ ≈120 µA before margin).

• 3.3/5 V push-pull: bias open; select power only if the board powers that sensor. Always connect signal ground.

• 3.3 V open drain that is not 5 V tolerant: bias open, external pullup to the sensor's own 3.3 V rail. There is no onboard 3.3 V sensor supply selector.

Design targets: 0-30 V positive input, LOW ≤1.2 V / HIGH ≥2.4 V, common-ground offset budget ±0.1 V. Do not use as an analog input, quadrature encoder, IO-Link port or continuous negative-voltage input. Sampling is 1 ms with 1-100 ms debounce; short pulses can be missed. Sensor power budgets are shared, 300 mA per rail. No rail voltage or fault telemetry is present.

Programming contacts

J8 uses the **TC2050-IDC-NL mechanical pattern and a custom electrical mapping**, centered at (24, 5.75) mm. Use a mapped adapter, not an assumed ARM/STDC ribbon pinout. Ten 0.7874 mm contact pads are on 1.27 mm pitch, with three mechanical guide holes. In the component-side view, the upper row runs **10, 9, 8, 7, 6** left to right at Y=5.115 mm; the lower row runs **1, 2, 3, 4, 5** at Y=6.385 mm. Pin 1 is at X=21.46 mm, Y=6.385 mm. Use this U-shaped numbering, not an odd/even ribbon numbering assumption.

J8 pin	Signal	Connection
1	NRST	Probe reset, optional
2	MCU UART RX	3.3 V UART adapter TX
3	MCU UART TX	UART adapter RX
4	SWCLK	SWD clock
5	SWDIO	SWD data
6	+3V3	Probe VTref sense **only**, board output
7	GND	Probe ground
8	GND	UART ground
9	BOOT0	Normally leave open; onboard pulldown
10	GND	Ground

Power the board through J1 during programming. Leave adapter/probe power disconnected. BOOT0 is normally low. SWD programming and custom pogo-adapter fit require bench verification. The intended clip-free probe is held perpendicular by a fixture; the no-legs pattern does not retain itself.

4 Hand assembly

Source: docs/ASSEMBLY.md

Use the exact BOM, assembly drawing, interactive BOM and placement files. Reference labels are in these drawings; dense placement does not leave room to print every reference legibly on the actual top silkscreen.

1. Order the 52 ×56 mm, two-layer, 1.6 mm board with 1 oz finished copper target. Use the exported Gerbers and separate plated/nonplated drill files. Have the fabricator review its actual stackup, annular rings, mask registration and minimum features. Gold finish is useful at the programming contacts; keep solder and paste off them.

2. Check bare-board continuity and orientation. Apply paste to SMT pads only. A 0.10 mm stencil is a starting point; the assembler must review release and the exposed-pad aperture. Solder selectors, TC2050 contacts and the test point deliberately have no paste.

3. Place 0805/1206 passives, diodes and ICs. Verify C1 polarity, TVS/diode cathodes, pin 1 on each IC and the exact latch-off TPS2553DBVR-1 suffix. U3 TPS1H200A has an exposed thermal pad that must be soldered to its ground land; use paste and a controlled heatbed/reflow profile. There are no components on the bottom.

4. Inspect joints and measure for supply shorts. Hand solder the six horizontal Weidmüller terminals after SMT reflow unless the assembler qualifies their THR process. Fill both plated holes per terminal pole. Support the board without loading fresh SMT joints. The terminal pins extend below the board.

5. Leave all selectors open for initial bring-up. Supply J1 from a current-limited bench supply, starting at 12 V and a modest current limit. Confirm input polarity, protected input, approximately 5.07 V internal and 3.3 V logic rails. Sensor headers remain unpowered while power selectors are open.

6. Program through the custom TC2050 mapping, verify flash and configure a unique CAN ID over UART. Then power off, choose each port's supply/bias and termination settings, inspect bridges and check that 5 V and battery are not shorted.

7. Perform the qualification checks before connecting a robot harness. Begin with a known load and known logic source, then the actual sensor/cable.

Use M3 hardware at (3.5,4) and (48.5,34.5) mm. Holes are 3.2 mm NPTH. The copper keepout has at least 3.30 mm radius; use hardware whose contact area stays within a 6.4 mm diameter, with insulating washers/spacers as appropriate. Account for component bodies, connector overhang and protruding pins. Do not clamp the PCB between conductive surfaces. The mechanical drawing gives nominal geometry, not a fitted enclosure.

The compact board still benefits from magnification and a fine iron. It is easier to assemble than B because it removes 55 fitted parts and all loose shunts, but it is not a through-hole beginner kit.

5 Firmware

Source: firmware/README.md

Bare-metal STM32F042K6T6 firmware for three digital inputs. Build allocation: 5,884 bytes flash, 1,808 bytes RAM including reserved stack; stack depth has not been measured. The final two 1 KiB flash pages hold redundant 32-byte versioned configuration with CRC and commit-last writes.

From `rev_c/`:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

Outputs: `build/sensor_can.elf`, `.hex`, `.bin`, `.map`. No ADC, sensor rail GPIO control or per-input LED logic remains. The 8 MHz external crystal and PLL generate 48 MHz; bxCAN uses 16 time quanta with an 87.5% sample point at 1 Mbit/s. Auto bus-off recovery is enabled; automatic transmit retry is disabled.

Power through J1 from a bench supply. Use the custom TC2050 wiring: VTref sense→6, SWDIO→5, SWCLK→4, ground→7, optional reset→1. Leave probe power disconnected.

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -c 'program firmware/build/sensor_can.elf verify reset exit'
```

This command has not been exercised on physical Rev C hardware. Verify MCU identity, supply and flash readback. Normal application programming should preserve the two configuration pages; full-chip erase removes them. Version-1 configuration is rejected.

UART: adapter TX→J8.2, RX→J8.3, ground→J8.8; 3.3 V, 115200 baud, 8N1; adapter power disconnected.

```
python host/sensor_can.py info --port /dev/ttyUSB0
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2
```

Device IDs 0-63 must be unique in this Team Use address space. Invert is a 3-bit mask; debounce is 1-100 ms. UART `SET id invert debounce_ms` persists and resets; CAN configuration is volatile. Unconfigured boards stay CAN-silent. **Hardware sensor power is enabled independently of firmware**, including during reset or programming; open power selectors disconnect individual ports.

The independent watchdog is serviced only after the full 1 ms foreground cycle. A >20 ms foreground delay, clock failure, hard fault or PVD low-voltage detection puts CAN into standby; the watchdog provides reset recovery. The single LED pulses approximately 100 ms per second when configured and healthy, or 100 ms per 200 ms for configuration/clock/CAN errors. It does not indicate sensor rail health.

Broadcast Disable ID 0 stops telemetry and aborts pending transmissions. Local sampling and hardware sensor power continue. An explicit `RESUME` UART command or CAN resume command re-enables telemetry; reads never do this automatically. See protocol v2. Host tests cover firmware logic and mocked peripheral boundaries, not actual oscillator startup, electrical timing or STM32 peripheral behavior.

6 CAN protocol

Source: docs/CAN_PROTOCOL.md

1 Mbit/s classic CAN, extended 29-bit identifiers, 8-byte telemetry. This is an original Team Use protocol, not CTRE Phoenix. Device type=10 (Miscellaneous), manufacturer=8 (Team Use), unique device ID=0-63. Address = `(10 << 24) | (8 << 16) | (API << 6) | device`.

The APIs differ from A/B/B-2 to prevent their rail-monitoring drivers from interpreting Lite frames. Provision each board over 3.3 V UART before connecting multiple boards. An unconfigured board remains CAN-silent; its sensor power hardware is still enabled.

All multibyte fields are little endian. Masks use bits 0-2 for inputs 1-3. Unsupported versions, reserved bits, wrong lengths or stale frames must be reported as unavailable.

API	Period	Eight-byte payload
0x120 status	10 ms	version=2, sequence, debounced input mask, raw mask, channels=3, capabilities=0x1F, flags, reset flags
0x121 counts 1/2	100 ms	two uint32 rising-edge totals
0x122 count 3	100 ms	uint32 total, uint32 reserved=0
0x130 diagnostics	100 ms	uint32 transmit drops, TEC byte, REC byte, LEC byte, CAN flags byte
0x131 identity	1000 ms	2, firmware major=0, minor=3, patch=0, hardware=3, device ID, debounce ms, inversion mask

Status flags: configured=1, clock healthy=2, CAN initialized=4, bus-off=8. Reset flags: power-on=1, pin reset=2, watchdog=4, software=8. Capabilities bits 0-4 indicate edge counts, runtime configuration, UART persistence, standalone hardware power and broadcast disable. They are **not live power health**. The raw mask is the uninverted sampled comparator state. The debounced mask applies configured logical inversion and debounce. Sequence is an unsigned byte and wraps. Counters wrap at 2^32 and clear on reset; runtime configuration preserves totals.

Diagnostic LEC is the STM32 three-bit last-error code (0-7); CAN flags are bus-off=1 and error-passive=2. There is no sensor voltage, current or rail-fault telemetry. An absent sensor supply can appear as an ordinary input state.

Command API	Payload	Effect
0x220 runtime input config	`53 43 02 invert debounce_lo debounce_hi 00 00`	invert 0-7; debounce 1-100 ms; volatile, does not change ID
0x221 resume telemetry	`53 43 02 00 00 00 00 00`	Explicitly resume after broadcast disable

Extended arbitration ID **0** is the FRC broadcast disable. Firmware stops telemetry and aborts pending transmit mailboxes immediately on processing it. Sampling and standalone sensor power continue; there are no actuator outputs. Resume requires the explicit command above, UART `RESUME`, or reset. Drivers never resume from a read loop. Integrate explicit resume with a reviewed robot lifecycle policy; this implementation still requires robot-bus testing and does not establish competition approval. See WPILib CAN addressing.

Read timeouts: status 100 ms, counters/diagnostics 350 ms, identity 1500 ms. Counters 1/2 and 3 are separate snapshots. Sampling is 1 kHz; debounce requires a continuously stable logical state for the selected interval. This is not a high-speed pulse counter or encoder.

UART: 115200 baud, 8N1, 3.3 V logic. Commands `INFO`, `HELP`, `SET id invert debounce_ms`, `RESUME`. `SET` saves a version-2 redundant CRC-protected record and resets. The old four-argument command and old flash record are rejected. No power-control command exists.

7 Robot integration

Source: examples/README.md

These drivers are for **protocol v2 / hardware 3 only**. Do not use the A/B/B-2 driver. Java classes are named `SensorCanLite` / `SensorCanLiteCodec`; C++ uses the `sensorcanlite` namespace. They compile against WPILib 2026.2.1. Python UART/SocketCAN tools are under `host/`.

Copy both Java files from `java/src/main/java/frc/robot/sensors/` into the matching package in a robot project. Construct the driver once, with the unique ID configured over UART:

```
private final SensorCanLite sensor = new SensorCanLite(17);

// In periodic(): inspect a fresh sample; empty means unavailable.
var sample = sensor.status();
if (sample.isPresent()) {
    var s = sample.get().value();
    if (s.configured() && s.clockHealthy() && s.canHealthy()) {
        boolean signal1 = s.input(1);
        // Display/use this signal with the application's sensor-fault policy.
    }
}
```

C++: copy `cpp/include/SensorCanLite*.h`; see the compiled `cpp/ReadSensorExample.cpp`. Read methods return `std::optional`; Java returns `Optional`. A missing, stale or malformed frame is unavailable, not false. Read timeout is 100 ms for status, 350 ms for counts/diagnostics and 1500 ms for identity.

A valid CAN status does not prove the sensor has power. Rev C intentionally removed rail monitoring. An unpowered sensor or latched rail can look inactive. Applications that move actuators must account for sensor supply/wiring faults independently. These examples only acquire input signals.

Runtime inversion/debounce changes are explicit and volatile. After broadcast disable, call `resumeTelemetry()` / `ResumeTelemetry()` only as an explicit reviewed lifecycle action; never continuously from `periodic()`. No driver read sends packets or changes sensor power. Sensor power cannot be switched through this protocol.

```
# Provision using a mapped 3.3 V UART adapter; install pyserial first.
python host/sensor_can.py configure --port /dev/ttyUSB0 --device 17 --invert 1 --debounce 2
# Passive monitor on an already configured 1 Mbit/s SocketCAN interface; requires python-can.
python host/sensor_can.py monitor --channel can0 --device 17
```

`bash scripts/test_drivers.sh` compiles Java/C++ against cached, checksum-pinned WPILib dependencies and runs host tests. Fetch dependencies with `python tools/fetch_driver_dependencies.py`. Real roboRIO/CAN testing remains pending.

8 Purchasing and factory quote

Source: docs/FACTORY_ORDER.md

Supply Factory_BOM.csv, SMT positions, all fitted positions, the fabrication directory, schematic, assembly drawings and this guide. A factory upload ZIP collects them. This is a quotation/DFM packet, not an order already placed or a production-qualified release.

• 52 ×56 mm, 2-layer, 1.6 mm FR-4; nominal 1 oz copper. Quote ENIG for clean pogo contacts, solder mask and both silkscreens. Panel rails/fiducials/tooling are assembler-owned; do not move individual components or alter nets.

• **89 placements: 83 SMT plus six Weidmüller THT/THR terminals**, all inserted from the top. No purchased jumpers or shunts. Keep the seven solder selectors open unless a specific configuration is supplied in writing.

• Quote exact manufacturer part numbers. DigiKey stock/price observations are dated; the factory can source through authorized channels. No unverified LCSC code or promised JLCPCB basic-library status is supplied.

• Generic resistors/capacitors may have price advantages locally, but match value, tolerance, power, temperature coefficient, voltage, dielectric, DC-bias performance and package. Preserve the precision 5 V divider, 1206 pulldowns and three specific output capacitors unless reviewed.

• Do not silently substitute TPS2553 without the **-1 latch-off** behavior, the MCU package, CAN transceiver, crystal load, buck variant/operating mode, inductor family, or Weidmüller connector angle/footprint.

• Weidmüller 1824420000/1824430000 have horizontal /90 entry, 3.5 mm pole pitch, two legs per pole, 7 mm row spacing and 1.1 mm PCB drills. /135 and Phoenix PTSM are not drop-in substitutions.

• Review pick-and-place rotations against pin-one drawings, U3 exposed-pad soldering, connector polarity, stencil apertures and terminal solder fill. KiCad coordinates use X right, Y up from the top-left design origin; do not infer a factory-specific rotation convention without review.

• Factory programming, continuity, functional test, attrition and terminal hand soldering should be quoted explicitly. Use supplied firmware and a custom mapped TC2050 fixture. Unique device IDs are provisioned later over UART.

The **$43.90** figure is a dated quantity-one component estimate, not an assembled-board price. PCB, SMT setup, stencil, THT labor, programming/test, shipping, tax, component reels/attrition and MOQ must be added by a real quote. Fewer placements should reduce assembly work versus B/B-2, but no China factory quote has been obtained. Keeping specified terminals is likely to limit savings more than replacing a few inexpensive 0805 parts.

9 Validation and qualification

Source: docs/VALIDATION.md

Unbuilt prototype. These are computational design checks. No physical power, sensor, CAN, thermal, EMI, ESD, pogo or assembly test has been performed.

Completed evidence

• Native KiCad 10.0.6 ERC/DRC, zero reported violations and zero unrouted connections: ERC, DRC. Reports list ignored optional checks; not every possible KiCad check is enabled.

• Independent component/pin/net comparison between explicit design, exported schematic and PCB: parity.

• Gerbonara comparison of Gerber pad identities/nets/coordinates, all drills, BOM and placements: export verification.

• Shapely copper-union/barrel graph finds no opens, shorts or unassigned islands on 70 identified nets: connectivity. It does not model defective plating or impedance.

• Exported copper edge/net/mount clearance, annular rings, hole spacing and mask/paste geometry: geometry.

• Official Weidmüller CAD/pin alignment checked for six connectors; native model references resolve locally.

• Firmware compiles with warnings as errors. Sanitized host tests exercise CRC/torn configuration, debounce/inversion, counter/timer rollover, malformed commands, configuration/clock/CAN failure, broadcast disable, UART overflow and watchdog scheduling: firmware tests.

• Java tests include stale/malformed data and no transmissions from reads; Python/C++ check versions, channel/reserved fields and commands. Actual WPILib 2026.2.1 transport headers/jars compile: driver tests.

Electrical budgets

Full formulas and assertions: check_electrical.py, results. Signal resistor budget is 1.75% including tolerance, drift and extra allowance. Logic rail ±3%; comparator offset allowance ±5.5 mV; per-input bias ±50 nA; output leakage 100 nA. TI specifies typical, not guaranteed maximum, input bias/leakage. These assumptions require temperature/lot measurements.

The 4,096 evaluated threshold extrema give rising 1.665-2.272 V and falling 1.401-1.933 V. Targets LOW ≤1.2 V / HIGH ≥2.4 V include ±0.1 V ground offset within the model. Nominal hysteresis is about 0.31 V; filter time constant about 18 µs. Sense voltage stays below 2.59 V at 30 V input. The 220 kΩ series resistor dissipates about 4 mW; the PNP pulldown about 92 mW worst budget, so it stays 1206 /0.25 W.

The precision buck divider targets 5.067 V. Reference/divider/leakage budget is 4.953-5.182 V; separate ±40 mV regulation allowance extends it to 4.913-5.222 V. After conservative load-switch drop, the lower 5 V rail budget is **4.871 V before PCB/connector/wire losses**. This does not establish 5 V ±5% at a remote sensor. Three 22 µF output parts give approximately 25.9 µF under typical DC-bias and extra budgets, not a guaranteed minimum.

At 6 V input, 300 mA per sensor rail plus pullups and 100 mA logic allowance, assumed 80% buck efficiency and 0.276 Ω entry resistance give approximately 0.773 A input and 5.787 V protected input. Duty budget is below 95%. Measure actual fuse/FET/trace loss, efficiency, dropout and ripple. Below 6 V the regulated sensor supply is not promised. The 3.3 V LDO loss allowance is about 0.192 W; thermal rise is unmeasured.

Limiter budgets are approximately 0.403-0.626 A battery and 0.394-0.506 A 5 V. They protect **shared rails**, not individual ports. Battery latch delay is roughly 11 ms typical. A heavily loaded rail with capacitive sensors can trip at startup: 40 µF at 22 V with only 83 mA charging margin already takes about 10.6 ms, leaving little margin. No arbitrary external-capacitance rating is established. Characterize actual inrush and delay extremes.

SMBJ22A standoff is 22 V, specified pulse clamp 35.5 V. Buck absolute maximum is 40 V; main bulk capacitor 50 V. Wiring inductance/overshoot can consume the margin; TVS protection is not sustained overvoltage regulation or load-dump qualification. Input SMAJ30A parts are transient clamps, not a continuous negative-voltage rating. CAN protection/common ground do not provide isolation.

Required bench work

Test	Exercise and record
Assembly / mechanics	Connector orientation/fill, U3 exposed pad, holes/washers, button access, pogo mapping and fixture fit
Power	6/12/22 V no/full shared loads; simultaneous startup; inrush/capacitance sweep; rail voltages, ripple, dropout, efficiency
Faults	Separate/simultaneous rail shorts; power-cycle recovery; protected reverse-input test; sensor backfeed
Temperature	FET, fuse, buck, inductor, LDO, limiters and pulldowns at load/ambient extremes; derate to measurements
Inputs	NPN/PNP and 3.3/5 V push-pull, OFF leakage, LOW residual, thresholds, hysteresis, cable offsets, pulse/debounce limits
CAN / firmware	Robot bus loading, bus-off, unplug/replug, broadcast disable/resume, unique IDs, brownout/HSE faults, watchdog
Disturbance	Conducted transient, ESD and EMC testing; actual peak clamps and recovery
Robot	Sensor supply through battery sag, stale/unpowered-sensor handling, WPILib integration and applicable season rules

Production qualification and FRC approval have not been established. No sensor supply health detection remains in Rev C.

10 Connector evidence

Source: docs/CONNECTORS.md

References	Exact MPN	Series description	DigiKey
J1-J3	**1824420000**	LSF-SMT 3.50/02/90 3.5SN BK TU	281-2020-ND
J4-J6	**1824430000**	LSF-SMT 3.50/03/90 3.5SN BK TU	281-2021-ND

Manufacturer pages: 2-pole, 3-pole. `/90` gives horizontal wire entry parallel to the PCB. `/135` is the angled version and is not used. Phoenix PTSM is not fitted.

These are through-hole/THR parts despite “SMT” in the family name. Pitch is 3.5 mm, with two solder legs per pole and 7 mm row spacing. PCB drills are 1.1 mm. Housing widths are 7.7 mm / 11.2 mm. Manufacturer CAD is 14.3 mm deep; the footprint conservatively reserves 14.75 mm. Height is 8.5 mm above the mounting surface; nominal pin tips reach 3.5 mm below that surface (about 1.9 mm beyond a 1.6 mm PCB).

Current renders use the official STEP solids with a rigid translation (-2.13,-12.625,-3.5) mm into the footprint origin. They are not approximate connector boxes. Display colors are assigned by the project. Original files, transformed files, URLs, hashes, pin/drill checks and nominal bounds are in vendor_connectors.json. Script `import_connector_models.py` checks all six placed pad patterns and the two vendor models. Do not regenerate models merely to change routing: STEP exporters can alter file headers without changing geometry.

Top terminals face outward at the power/CAN edge; three sensor terminals face outward at the opposite edge. Conservative complete footprint envelope is 52 ×64.1 mm; add room for cable bend, strain relief and button access. Renders do not validate conductor retention, tool clearance in an enclosure, molding tolerances or actual solder fit.

11 Tools and reproduction

Source: TOOLS.md

The saved native KiCad schematic/PCB are authoritative. Generation/placement/router scripts record design construction; they are not a one-command reroute of the reviewed board. Editing the circuit requires matching schematic, BOM, PCB and new checks. Earlier revisions remain separate.

Tested toolchain: KiCad 10.0.6, its matching Python/pcbnew interpreter, Python 3.12, Arm GNU GCC 13.2.1, GNU make, host GCC/G++, Java 17, Poppler, InteractiveHtmlBom 2.11.2 and the packages pinned in `tools-requirements.lock`. Tool binaries/caches are external to the repository. Install those native tools for the host operating system. The `kicad-python` wrapper must use the interpreter matching the KiCad binary bindings, not an arbitrary venv Python.

```
# In rev_c/, after installing native tools:
uv venv --python python3.12 .venv
uv pip sync --python .venv/bin/python tools-requirements.lock
source tools/env.sh
python tools/fetch_driver_dependencies.py
bash scripts/validate_release.sh
```

`tools/env.sh` uses a revision-local venv, or the repository venv when working in the full checkout. It prepends `$CANDI_TOOLS_ROOT/bin` (default `$HOME/.local/share/candi-tools/bin`). Wrappers expected there are `kicad-cli`, `kicad-python`, `host-gcc`, `host-g++`, `ibom`, and Arm compiler commands. Java is under the configured tools root or selected with `CANDI_JDK`. Fonts for the PDFs are DejaVu Sans/Mono.

The validation script checks the saved design and refreshes manufacturing evidence, firmware and host tests; it does not move or reroute components. A visual/documentation refresh is separate:

```
bash scripts/render_board.sh
python scripts/build_documents.py
python scripts/build_handbook.py
python scripts/review_documents.py
python scripts/build_release.py
```

The factory packet is built by `build_documents.py`. Rebuild it after any BOM/placement/fabrication change. `build_release.py` selects authoritative files and writes a manifest plus ZIP checksum. Check an extracted archive with `verify_archive.py`; the release manifest deliberately excludes itself and the enclosing ZIP from its file hashes.

Use `python scripts/import_connector_models.py` only when restoring/reviewing vendor models. It checks the original STEP hashes and all six native connector pad patterns, then writes translated local models. It can change STEP headers, requiring a fresh STEP export, renders and release. Keep the originals and provenance JSON together.

`hardware/design/circuit.py` is the explicit component/net model; `generate_schematic.py` rebuilds the schematic. Construction helpers such as `pack_compact.py`, `route_critical.py` and `apply_routes.py` operate on intermediate files and can overwrite placement/routing. They are not part of release validation. No SPICE transient or thermal qualification has been added for Rev C; do not reuse older revision reports as evidence for this changed supply.

12 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

KiCad library/model notices are retained under `licenses/`, together with Arm CMSIS, ST device-header and WPILib notices. Firmware third-party directories and InteractiveHtmlBom output retain their notices. Original project work has no selected blanket license. Manufacturer documents/CAD remain their owners' material; retaining a file is not a redistribution-license grant.

Subject	Primary reference
Buck and capacitors	TI LMR51420
Battery limiter, IN tied to VS standalone mode	TI TPS1H200A-Q1, section 6.3.3 in reviewed Rev E
5 V latch-off limiter	TI TPS2553, -1 variant
Comparator/leakage caveats	TI LM339LV
CAN interface / bypass	TI TCAN3413
STM32 pinout, supplies, clock	ST STM32F042
Example industrial output residual levels	Omron E3Z specifications
Terminal geometry and exact MPNs	Weidmüller 1824420000, 1824430000
CAN addressing / broadcast disable	WPILib

Samsung CL31A226KAHNNNE typical characteristic data is retained in `sources/datasheets/samsung_CL31A226_characteristics.json`, inherited from the earlier design's manufacturer retrieval. Typical curves do not establish guaranteed capacitance. DigiKey URLs, exact codes and dated observations are in `bom/catalog_selected.json`. Vendor STEP URLs, hashes and transforms are in `hardware/library/3d/vendor_connectors.json`.

Equivalent parts need engineering review; distributor substitution suggestions do not establish electrical/mechanical equivalence. Availability and season rules can change after these observations.